

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AMA574
Subject Title	Generative AI and Foundation Models
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	This course aims to equip students with a comprehensive understanding of the mathematical foundations and applications of Generative Artificial Intelligence (AI) and Foundation Models. It covers the mathematical foundations, model architectures, and training methodologies behind key generative models, including Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), diffusion models, Mixture-of-Experts (MoE) models, and MAMBA. The course also explores large-scale Transformer-based architectures such as GPT and BERT. Students will examine the practical applications of these models across various domains, gaining insights into their capabilities, limitations, and broader societal implications. Ethical considerations, responsible deployment, and emerging trends in generative AI development will also be discussed.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) explain the mathematical foundations and training principles underlying generative and foundation models, including concepts from probability, optimization, and representation learning; b) describe and compare major generative model architectures such as GANs, VAEs, diffusion models, Mixture-of-Experts (MoE), and MAMBA, as well as large-scale Transformer-based models such as GPT and BERT; c) analyze the design, training strategies, and application scenarios of generative AI systems across different domains; d) evaluate model performance using concepts such as generalization, capacity, overfitting, underfitting, and regularization; e) assess the limitations, ethical challenges, and societal impacts of deploying generative and foundation models in real-world settings.
Subject Synopsis/ Indicative Syllabus	Key topics to be delivered: a) Introduction to Generative AI and Foundation Models: Historical development and motivation; fundamental concepts and definitions; key milestones in generative AI research. b) Core Generative Architectures: Describe and compare major generative model architectures, including VAEs, GANs, diffusion

	models, Mixture-of-Experts (MoE), and Mamba, as well as large-scale Transformer-based models such as GPT and BERT. c) Applications and Case Studies: Real-world use cases across domains: text generation (e.g., chatbots, content creation), image synthesis, video generation, and multimodal models. Industrial deployment scenarios and emerging applications will also be examined. d) Ethical, Societal, and Future Perspectives: Discussion of key ethical challenges, including fairness, bias, misinformation, and potential misuse. Students will explore regulatory considerations, responsible AI development, and future trends in generative and foundation model research.																																													
Teaching/Learning Methodology	<u>Lectures:</u> Lectures will introduce foundational theories, model architectures, and key technical results. Mathematical derivations and real-world examples will be used to illustrate concepts. <u>Tutorials:</u> Tutorials will focus on hands-on practice, including coding exercises, model training, and group discussions of recent case studies and research papers.																																													
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr> </thead> <tbody> <tr> <td>1. Project</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>2. Assignments</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr> <tr> <td>3. Exam</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>Total</td><td>100%</td><td colspan="5"></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: <u>Project / Assignments:</u> Students will design, implement, and analyze a generative model or foundation model on a chosen dataset or task, demonstrating practical skills and theoretical understanding. <u>Exam</u> A written exam will test conceptual understanding, ability to analyze model behaviors, and evaluate broader impacts of generative AI technologies.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Project	30%	✓	✓	✓	✓	✓	2. Assignments	20%	✓	✓	✓	✓		3. Exam	50%	✓	✓	✓	✓	✓	Total	100%					
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3. Exam	50%	✓	✓	✓	✓	✓																																								
Total	100%																																													
Student Study Effort Expected	Class contact:																																													
	▪ Lecture				26 Hrs.																																									
	▪ Tutorial				13 Hrs.																																									
	Other student study effort:																																													

	▪ Self-learning from reference materials	36 Hrs.
	▪ Reading and research for the case study	42 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	[1] Goodfellow, I., Bengio, Y., & Courville, A., <i>Deep Learning</i>, MIT Press, 2016 (Chapters on generative models). [2] Kingma, D. P., & Welling, M., "Auto-Encoding Variational Bayes," ICLR, 2014. [3] Radford, A., et al., "Language Models are Few-Shot Learners," NeurIPS, 2020 (GPT-3 paper). [4] Ho, J., et al., "Denoising Diffusion Probabilistic Models," NeurIPS, 2020. [5] Bommasani, R., et al., "On the Opportunities and Risks of Foundation Models," Stanford CRFM Report, 2021.	

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